

## Python interview questions with answer

1. 1<sup>st</sup> version of python ? who invented ? and inventend in which year?

Ans- python was developed by Van Guido Rossum in late 1980' and early 1990's at the national research mathematics and computer science netherland. (first version appeared in 20 feb 1991- 0.9 .o or v1.0 or 1.x) now 3.10.2 worked 3.9.0

2. Differences in between list , tuple,set, dictioanary, ,string , float.

3. Difference between data types and data structure.- data store in memory can many types

| Dynamic data type                               | Standard/ Structure data types- structuring data in particular format called data structure. |
|-------------------------------------------------|----------------------------------------------------------------------------------------------|
| Numeric                                         |                                                                                              |
| 1. Integer- complete no (100)                   | 1. List [ ]                                                                                  |
| 2. Float – incomplete no 12.5                   | 2. Tuple ( )                                                                                 |
| 3. Complex – real no+img no (10+11j)            | 3. Set { }                                                                                   |
| 4. Boolein- 1. true 2. false                    | 4. Dictionary { }                                                                            |
| 5. String – group of character (not in python ) | 5. String ''                                                                                 |

4. what is difference between append and extend?

**Append**- single element insert at the end in list a.append(20)

**Extend**- multiple element insert at the end in list a.extend([20,30])

5. can we define list inside tuple ? ans – **yes** - [(1,2,3), ('S', 'P', 'Q')]

6. can we define tuple insdie list? Ans - no because tuple is immutable

7. why delete random elements from set in python ?

8. what is difference between displaying and print?

**Print**- The print function is applied to represent any kind of information like string or estimated value.

**Display** - The displaying function is a built-in feature of the Python language and utilized to understand a line of a text provided by the user. Eg – print ('rushi') print (2+3) , 2+3 = 5 display a = 500 , b = 'rushi'

9. in dictionary what is key and value?

Keys – keys are most commpnly string, number or tuble or we can use immuatable data types.

Values- values can be mutable or immutable. Can be any type that you want.

10. Can access the values by using keys or can be assign list as a key ? -

**no** because for every key it generates a [hash value](#) for the key and by this hash value, it can track its elements.

11. Can we add duplicate value pair in dictionary?

NO – In dictionary keys always unique if we are try to give duplicate key then it will be error.

12. What is doc's string or documented string ? vimp for experienced candidate

Doc's sting is additional information about code (jenva developer code lihto to samorchyala samjla pahije tya sathi use hoto)

13. Difference between loop control statement , continue and break ?

1. define a for loop to add elements in list.

**Loop control statements**- Loop control statements are **used to change the flow of execution**. These can be used if you wish to skip an iteration or stop the execution. Types – break,continue ,pass

**break** - leaves the loop, skips remaining execution of complete loop, useful, if condition always evaluates to be True.

**continue**- jumps for next iteration, skips execution of remaining statement(s) inside the loop for current iteration, useful, if wants to skip execution of some statement(s) inside the loop for any particular iteration

```
For l in range (1,11):
    If i>=5;
        Break
Print (i)
1,2,3,4,5
```

```
For l in range (1,11):
    If i>=5;
        continue
Print (i)
1,2,3,4,,6,7,8,9,10,
```

```
a=[]
for i in range(5):
    a.append(i)
print(a)
# [0, 1, 2, 3, 4]
```

14. **Assignment operator = Assignment operators** are used in **Python** to assign values to variables. a = 5 is a simple **assignment operator** that assigns the ...

15. What is pep8 ?- it is used in positional argument function and having minimum length of arg is 8.

16. What is \* arg and \*\*karg –

- **\*arg** – argument variable stored in tuple. Def details (name,age,salary) , print (f" name:{name}")
- **\*\*kwarg** - key or values stored in dictionary. Def details (name='rushi', age= 26)

17. What is list comprehension ? – for compress the code. List comprehension is generally more compact and faster than normal functions and loops for creating list.

|                                                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 18. Tuple comprehension is possible ? – no , if we create tuple comprehension . it will generate generator.                                                                                                                             |
| 19. Dictionary comprehension is possible? – yes .                                                                                                                                                                                       |
| 20. What is lambda function? – it is an anonymous function or no name , or one line function or higher order func . we need this function for temporary task or just in time. == (lambda x,y :x+y )(2,3) or x = (lambda x,y :x+y )(2,3) |
| 21. What is map ? – it is used instead of loop. It is the function which takes function as argument and applies on sequence .direct square hoat yaat . a= (3,4,5) . list (map(lambda x:x*x,a)) or list(map(sqr,a)) o/p [9,16,25]        |
| 22. Print ? - The print() function prints the specified message to the screen, or other standard output device. The message can be a string, or any other object,                                                                       |

### Advance python

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| 23. What is self? –<br><ol style="list-style-type: none"> <li>1. self represents the instance of the class. By using the “self” keyword we can access the attributes and methods of the class in python. Or self variables which refers current class instance class.</li> <li>2. It binds the attributes with the given arguments. We can create multiple of a class and each instance will have different values ...</li> </ol>                                                                                           |
| 24. What is __init__ or use of __init__<br><ol style="list-style-type: none"> <li>1. It is as like constructor, associate with data.</li> <li>2. Or it is used to initialize the variables.</li> </ol>                                                                                                                                                                                                                                                                                                                      |
| 25. How many types of internal decorator? – where used decorator in ur project??<br><b>Decorators</b> are a very powerful and useful tool in <b>Python</b> since it allows programmers to modify the behaviour of function or class                                                                                                                                                                                                                                                                                         |
| 26. What is MRO? Method Resolution Order-                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 27. What is super method ? and use of super method?<br><ul style="list-style-type: none"> <li>- By using super method , it gives the access to parent class and sibling class.(super method ni aapan parent class la ani sibling la call kru shhkto)</li> </ul>                                                                                                                                                                                                                                                             |
| 28. Can we implement overloading in python ?<br><ul style="list-style-type: none"> <li>- Yes . we can implement. By using variable length or default argument or it is used to call parent class constructor or methods from child class.</li> </ul>                                                                                                                                                                                                                                                                        |
| 29. How we can achieve the abstraction ? have u worked on any abstract class? how we create it?<br><pre>from abc import ABC,</pre> <ul style="list-style-type: none"> <li>- An <b>abstract</b> class <b>can</b> be considered as a blueprint for other classes. It allows <b>you</b> to create a set of methods that must be created within .</li> <li>- It is use for security purpose or no object created for security , hide process in between input and outpur.</li> </ul>                                            |
| 30. Access specifier- Access specifiers or access modifiers in python programming are <b>used to limit the access of class variables and class methods outside of class while implementing the concepts of inheritance</b> . This can be achieved by: Public, Private and Protected keyword.                                                                                                                                                                                                                                |
| 31. Weather python is secure or not outside class?                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 32. What is difference between normal directory and python package? what are modules and packages in python?<br><ul style="list-style-type: none"> <li>- A package is a <b>collection of Python modules</b>, i.e., a package is a directory of Python modules containing an additional <code>__init__.py</code> file. The <code>__init__.py</code> distinguishes a package from a directory that just happens to contain a bunch of Python script.</li> <li>- Single folder is called directory.</li> </ul>                 |
| 33. Why we are using withopen ?<br><ul style="list-style-type: none"> <li>- Automatic close the the file we don't have need to close it when we are using withopen.</li> </ul>                                                                                                                                                                                                                                                                                                                                              |
| 34. What is use of rays?- Using Ray, you can take Python code that runs sequentially and transform it into a distributed application with minimal code changes.(log file madhe kuthe issue aahe he shodhun kadhta)                                                                                                                                                                                                                                                                                                          |
| 35. What is warning and above ?- The default level is <b>WARNING</b> , which means that only events of this level and <b>above</b> will be tracked, unless the logging package is configured to do ...<br>Warning messages are typically issued in situations where it is useful to alert the user of some condition in a program, where that condition (normally) doesn't warrant raising an exception and terminating the program. For example, one might want to issue a warning when a program uses an obsolete module. |
| 36. What is difference dump() and dumps() ?<br>Dumps – python dictionary to json string<br>Dump – python dictionary to json string then convert into file.                                                                                                                                                                                                                                                                                                                                                                  |
| 37. What is use of join method ?<br><ul style="list-style-type: none"> <li>- is used <b>to join a thread with the main thread</b> i.e. when join() is used for a particular thread the main thread will stop executing until the execution of joined thread is complete.</li> </ul>                                                                                                                                                                                                                                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

### 38. difference between sort() and sorted()?

- In Python, sorting any sequence is very easy as it provides in-built methods for sorting. Two such methods are sorted() and sort(). These two methods are used for sorting but are quite different in their own way. Let's have a look at them one by one.

sorted() method sorts the given sequence as well as set and dictionary(which is not a sequence) either in ascending order or in descending order(does unicode comparison for string char by char) and always return the a sorted list. This method doesn't effect the original sequence.

### 39. what are generators?

- If we have big data then us genretor it is same as like range
- Py =( x for x in range (100000000000)) ----- py=[ x for x in range (10000000)] not possible may be hang
- We are generating single number at a point – it is technique to improve

### 40. Whenever we have yield value in function is called generator

### 41. What is use of yield ?

In its simplest form, a yield statement looks much like a return statement, except that instead of stopping execution of the function and returning, yield instead **provides a value to the code looping over the generator and pauses execution of the generator function.**

### 42. what is deep copy and shallow copy? difference between deep copy and shallow copy.

- Deep copy – create new copy with existing reference

|                                                                                               |                                                                                              |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| Shallow Copy stores the references of objects to the original memory address.                 | Deep copy stores copies of the object's value.                                               |
| Changes not impacted                                                                          | Changes impacted                                                                             |
| Shallow Copy stores the copy of the original object and points the references to the objects. | Deep copy stores the copy of the original object and recursively copies the objects as well. |
| Shallow copy is faster.                                                                       | Deep copy is comparatively slower                                                            |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
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| <p>2. <b>features of python languages.</b></p> <ol style="list-style-type: none"> <li>1. Easy to code: Python is a high-level programming language. ...</li> <li>2. Free and Open Source: ...</li> <li>3. Object-Oriented Language: ...</li> <li>4. GUI Programming Support: ...</li> <li>5. High-Level Language: ...</li> <li>6. Extensible feature: ...</li> <li>7. Python is Portable language: ...</li> <li>8. Python is Integrated language</li> </ol> |
| <p>3. <b>how memory is managed in python?</b></p> <p>As we know, Python uses the dynamic memory allocation which is managed by the Heap data structure. Memory Heap holds the objects and other data structures that will be used in the program. Python memory manager manages the allocation or de-allocation of the heap memory space through the API functions.</p>                                                                                     |
| <p>4. <b>how python code gets executed??</b></p> <p>In Python, the source code is compiled into a much simpler form called bytecode. These are instructions similar in spirit to CPU instructions, but instead of being executed by the CPU, they are executed <b>by software called a virtual machine.</b></p>                                                                                                                                             |
| <p>5. <b>programmetical difference between python and c languages</b></p> <p>6. <b>how memory management done in python?</b></p> <p><b>C is a general-purpose, procedural computer programming language. Python is an interpreted, high-level, general-purpose programming language.</b> Compiled programs execute faster as compared to interpreted programs. Interpreted programs execute slower as compared to compiled programs.</p>                    |
| <p>7. <b>what is python -- interpreted or compiled?</b></p> <p>Python is an <b>interpreted language</b>, which means the source code of a Python program is converted into bytecode that is then executed by the Python virtual machine. Python is different from major compiled languages, such as C and C + +, as Python code is not required to be built and linked like code for these languages.</p>                                                   |
| <p>8. <b>how python file gets compiled?</b></p> <p><b>Python</b> code, written in <b>.py file</b> is first <b>compiled</b> to what is called bytecode (discussed in ... When that number <b>goes</b> down to zero, it deletes that object.</p>                                                                                                                                                                                                              |
| <p>9. <b>what are closures in python?</b></p> <p>It can access these variables even after the outer function has completed its execution.</p>                                                                                                                                                                                                                                                                                                               |
| <p>10. <b>eval() keyword in python?</b></p> <p>In simple words, the eval function evaluates the “String” like a python expression and returns the result as an integer.</p>                                                                                                                                                                                                                                                                                 |
| <p>11. <b>what is slots in python?</b></p> <p><b>slots</b> provide a special mechanism to reduce the size of objects.It is a concept of memory optimisation on objects.</p>                                                                                                                                                                                                                                                                                 |
| <p>12. <b>how to delete a file frompython?—os.remove (file name/directory)</b></p>                                                                                                                                                                                                                                                                                                                                                                          |
| <p>13. <b>can programmer access private heap space? does it allow inpython?</b></p> <p><b>The programmer does not have access to this private heap.</b> The python interpreter takes care of this instead.</p>                                                                                                                                                                                                                                              |
| <p>14. <b>what types of scope in python?</b></p> <p>- Local Scope. The Variables which are defined in the function are a local scope of the variable. ...</p> <p>Global Scope. The Variable which can be read from anywhere in the program is known as a global scope. ...</p> <p>NonLocal or Enclosing Scope. ...</p> <p>Built-in Scope</p>                                                                                                                |
| <p>15. <b>what is outer-most scope in python?</b></p> <p>[..]all you can do is make a variable in a particular scope. (If you make a variable inside the Python interpreter, and then import other modules, <b>your variable</b> is in the outermost scope and thus global within your Python session</p>                                                                                                                                                   |
| <p>16. <b>what is module level scope inpython?</b></p> <p>Names in this Python scope are visible from everywhere in your code.</p>                                                                                                                                                                                                                                                                                                                          |
| <p>17. <b>why should I use namespace in python?</b></p> <p>Namespaces <b>help us uniquely identify all the names inside a program.</b></p>                                                                                                                                                                                                                                                                                                                  |
| <p>18. <b>what is scope resolution in python?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                       |
| <p>19. <b>what is callable() type in python?</b></p> <p>a callable is <b>something that can be called</b>. This built-in method in Python checks and returns True if the object passed appears to be callable, but may not be, otherwise False.</p>                                                                                                                                                                                                         |
| <p>20. <b>what is pass keyword?</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <p>21. <b>what datatypes we can use as a key?</b></p> <p>you can use an <b>integer, float, string, or Boolean</b> as a dictionary key.</p>                                                                                                                                                                                                                                                                                                                  |
| <p>22. <b>what is linked list?</b></p> <p>- <b>a sequence of data structures</b>, which are connected together via links. Linked List is a sequence of links which contains items</p>                                                                                                                                                                                                                                                                       |

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| 23. how to combine 2 list-- by using +operator (a= list1+list2)                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 24. what is split() in python? –<br>cut specific element and or output is in list format- a = 'rushu is smart' (a.split('rushu') = ['is' 'smart']                                                                                                                                                                                                                                                                                                                                     |
| 25. scenario question --- code for list and range                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 26. write a list comprehension for even numbers.<br>l=['ab',1,2,3] any(l) what will be output?                                                                                                                                                                                                                                                                                                                                                                                        |
| 27. what will be error if we trying to change/update tuple element?                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 28. difference between python array and list                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 29. 5 built-in functions in python<br>1. type( ) function    2. Input( ) function    3. abs( ) function    4. pow( ) function<br>5. dir( ) function    6.sorted( ) function    7. max( ) function    8. round( ) function<br>9. divmod( ) function    10. id( ) function    11. ord( ) function    12.len( ) function<br>13. sum( ) function    14.help( ) function                                                                                                                   |
| 30. what is type hints in python?                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 31. write down program to explain string slicing concept.<br>By using index number copy that specific element - ASTRING - S1=Slice(3) – ast -- start: end: stop<br>s2 = slice(1, 5, 2) - SR                                                                                                                                                                                                                                                                                           |
| 32. how to access tuple elements? – by using indexing                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 33. what is dictionary? (-                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 34. how to merge 2 dictionaries? A= {} b = {}, c= merge(a,b)                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 35. write a dictionary comprehension for even numbers upto 100.<br>keys = ['a','b','c','d','e']<br>values = [1,2,3,4,5]<br>myDict = { k:v for (k,v) in zip(keys, values) }                                                                                                                                                                                                                                                                                                            |
| 36. keys are mutable or immutable in dict? – immutable                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 37. str='abcd' print only 'abc'? how to do it?                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 38. suppose their is a string=('a','e','i','o','u') how to make it different like ('a','e','i'), ('o','u')?                                                                                                                                                                                                                                                                                                                                                                           |
| 39. range is function or data type?<br>it is an in-built <b>function</b> in Python which returns a sequence of numbers starting from 0 and increments to 1 until it reaches a specified number. The most common use of range function is to iterate sequence type. It is most commonly used in <b>for and while loops</b> .<br>range(start, stop, step)                                                                                                                               |
| 40. what is any() function? Or purpose of any() function?<br>The any() function <b>returns True if any item in an iterable are true, otherwise it returns False.</b>                                                                                                                                                                                                                                                                                                                  |
| 41. difference between classmethod, staticmethod, instancemethod?<br>1. The instance method acts on an object's attributes. It can modify the object state by changing the value of <b>instance variables</b> .<br>2. Static methods have limited use because they don't have access to the attributes of an object (instance variables) and class attributes (class variables).<br>3. Class method which act upon the class variables of the clasee by using decorator @ classmethod |
| 42. we have classmethod then why we need staticmethods?<br>A static method has two main purposes: ... Since <b>there is no need to access instance variables</b> , having static methods eliminates the need for the caller to instantiate the object just to call the method. For the state that is shared by all instances of the class, like a counter. All instance must share the same state.                                                                                    |

#### 43. what are different methods of writing python functions

1. Python Function with no argument and no return value.
2. Function with no argument and with a Return value.
3. Python Function with argument and No Return value.
4. Function with argument and return value.

#### 44. difference between variables static & instance variables

- A static variable is a property of a class. . A static variable is created only once when the classloader loads the class.
- An instance variable is a property of an instance . An instance variable is created everytime an instance is created.

#### 45. difference between static method & instance method

- Static methods can be called without the object of the class. Static methods are associated with the class.
- Instance methods require an object of the class. Instance methods are associated with the objects

#### 46. difference between class method and static method?

A class method takes cls as the first parameter while a static method needs no specific parameters. A class method can access or modify the class state while a static method can't access or modify it. In general, static methods know nothing about the class state

#### 47. in a function assigned a value x=10 then how it get stored in python?

#### 48. What is decorator and where it is use

Decorators are a very powerful and useful tool in Python since it **allows programmers to modify the behaviour of function or class**. Decorators allow us to wrap another function in order to extend the behaviour of the wrapped function, without permanently modifying it.

#### 49. What is the use of exception handling? how exceptions gets handled in python?

Exception handling **ensures that the flow of the program doesn't break when an exception occurs**. For example, if a program has bunch of statements and an exception occurs mid way after executing certain statements then the statements after the exception will not execute and the program will terminate abruptly.

#### 50. what is stackoverflow exception in python?

-The exception that is thrown when the exception stack overflows because it contains too many nested method calls. This class can not be inherited.

#### 51. how many block in exception handling or flow of exception handling?

1. try – it contains code which may cause exception.
2. except – it is used to catch exception that is raised in try block .
3. can be multiple except block for try block.
4. else- it will execute when no exception is raised .
5. finally – whether there is exception or not .

```
def divide(x, y):  
    try:  
        print(f'{x}/{y} is {x / y}')  
    except ZeroDivisionError as e:  
        print(e)  
    except TypeError as e:  
        print(e)  
    except ValueError as e:  
        print(e)
```

#### 52. how u work with polymorphism in python?

Polymorphism in python **defines methods in the child class that have the same name as the methods in the parent class**. In inheritance, the child class inherits the methods from the parent class. Also, it is possible to modify a method in a child class that it has inherited from the parent class

#### 53. what are common built-in data types? what is range?

1. Binary Types: memoryview, bytearray, bytes.
2. Boolean Type: bool
3. Set Types: frozenset, set.
4. Mapping Type: dict.
5. Sequence Types: range, tuple, list.
6. Numeric Types: complex, float, int.
7. Text Type: str.

#### 54. what is `__repr__` in class?

#### 55. difference between `.get` and `.filter` method – django

#### 56. what is ORM? – in django

#### 57. Service base company or product base company? – ask to interviewee in last